BB84: Worked Problems

Problem 1 — Sifted-key fraction

Setup. Alice and Bob each choose Z or X basis independently and uniformly at random. Alice sends N qubits.

Ask. What fraction of transmitted bits survive sifting (i.e., bases match)?

Solution.

- P(bases match) = P(ZZ) + P(XX) = $\frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$.
- **Sifted bits:** $N_{\text{sift}} = \frac{N}{2}$.

Problem 2 — QBER under intercept–resend (naïve Eve)

Setup. Eve measures each qubit in Z with prob 1/2 and X with prob 1/2, then resends the post-measurement state. Channel itself is noiseless.

Ask. What is the Quantum Bit Error Rate (QBER) on **sifted** bits?

Solution.
Consider only positions where Alice and Bob used the **same** basis (those are the sifted bits).

- Case A: Eve used the **same** basis (prob = 1/2): she forwards exactly Alice's bit → **0%** error in these.
- Case B: Eve used the **wrong** basis (prob = 1/2): her outcome is random; the state she resends is random *in Alice/Bob's basis* → Bob's bit mismatches Alice's with prob **1/2**.

Thus, conditioned on sifting,

$$\text{QBER} = (1/2) \cdot 0 + (1/2) \cdot (1/2) \;=\; \boxed{25\%}.$$

Exercise 2.9

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Step 1 — Error rate introduced by Eve

- If Eve performs the simple **intercept–resend attack** (measuring in a random basis and resending):
  - On sifted bits (Alice and Bob chose the same basis), the **Quantum Bit Error Rate (QBER)** is:

$$q = 25\% = 0.25$$

Step 2 — Probability of not detecting Eve

Suppose Alice and Bob compare s sifted bits.

- Each compared bit reveals an error with probability $q = 0.25$.
- Probability a single bit shows **no error** = $1 - q = 0.75$.
- Probability that **all s bits show no error** =

$$(0.75)^s$$

Step 3 — Condition for 90% detection

- Probability of detecting Eve = $1 - (0.75)^s$.
- We want:

$$1 - (0.75)^s \geq 0.90$$

$$(0.75)^s \leq 0.10$$

Step 4 — Solve for s

Take natural logs:

$$s \cdot \ln(0.75) \leq \ln(0.10)$$

$$s \geq \frac{\ln(0.10)}{\ln(0.75)}$$

- $\ln(0.10) \approx -2.3026$
- $\ln(0.75) \approx -0.2877$

$$s \geq \frac{-2.3026}{-0.2877} \approx 8.0$$

Alice and Bob need to compare **at least 8 sifted bits** to have a **90% chance** of detecting Eve's presence.

General formula:

If the sifted-key error rate due to Eve is q (for intercept–resend in BB84, $q = \frac{1}{4}$), and you want detection probability at least P_{det}, compare

$$\boxed{s \;\geq\; \left\lceil \frac{\ln(1 - P_{\text{det}})}{\ln(1 - q)} \right\rceil}.$$

For BB84's $q = 0.25$:

$$s \;\geq\; \left\lceil \frac{\ln(1 - P_{\text{det}})}{\ln(0.75)} \right\rceil.$$

Quick table (BB84, $q = 0.25$)

Target detection P_{det}	Required bits s
80%	6
90%	**9**
95%	11
99%	17
99.9%	25
99.99%	33

Note on the 90% row: with $s = 8$, $1 - 0.75^8 \approx 89.99\%$ (just under 90%), so to *guarantee* ≥90% you need **9** bits.

Exercise 2.10. Analyze Eve's success in eavesdropping on the BB84 protocol if she does not even know which two bases to choose from and so chooses a basis at random at each step.

a. On average, what percentage of bit values of the final key will Eve know for sure after listening to Alice and Bob's conversation on the public channel?

b. On average, what percentage of bits in her string are correct?

c. How many bits do Alice and Bob need to compare to have a 90 percent chance of detecting Eve's presence?

Exercise 2.10

Problem restated:

Eve is even weaker than in the usual intercept–resend. She doesn't know which *two* bases are possible, so she picks a measurement basis at random from the **entire set of possible bases** (not just Z or X). Then she resends the result.

We are asked:

(a) On average, what percentage of final key bits does Eve know *for sure* after Alice and Bob's basis reconciliation?

(b) On average, what percentage of bits in Eve's string are correct?

(c) How many bits must Alice and Bob compare to detect her with 90% probability?

Step 1 — What's happening here

- In normal BB84, Alice uses Z or X; Eve doesn't know which, but she *does* know the set {Z, X}.
- Here, Eve is even more ignorant: she doesn't know the basis set, so she just picks a *random basis from the entire Bloch sphere*.
- This means:
  - The probability she *accidentally* picks exactly the correct basis is **zero** (continuous set).
  - Therefore, she never has *certain* knowledge of any final key bit.

(a) Percentage of final key bits Eve knows for sure

- Since she can never align exactly with Alice's basis, **Eve knows 0% of the final key bits with certainty**.

  Answer: $\boxed{0\%}$

(b) Percentage of Eve's guesses that are correct

- Eve measures each qubit in some random basis.
- From her result and the public announcement (which basis was used), she can make a *guess*.
- But since her measurement basis was random, her outcome is completely uncorrelated with Alice's bit (when conditioned on knowing the correct basis afterward).
- That means Eve is right with probability **1/2**.

Answer: $\boxed{50\%}$

(c) Detection probability for Alice and Bob

Now consider Bob's bits:

- If Bob uses the correct basis, then without Eve he matches Alice perfectly.
- With Eve interfering:
  - The state Bob receives is Eve's post-measurement state, which in Alice's basis looks completely random.
  - So Bob's bit agrees with Alice's with probability **1/2**, i.e. QBER = 50%.

Now, suppose Alice and Bob sacrifice s bits of the sifted key to test for errors:

- Each test bit is wrong with probability 0.5.
- Probability all test bits are correct = $(0.5)^s$.
- Probability they detect Eve = $1 - (0.5)^s$.

For 90% detection:

$$1 - (0.5)^s \geq 0.9 \quad \Rightarrow \quad (0.5)^s \leq 0.1$$

$$s \geq \frac{\ln(0.1)}{\ln(0.5)} \approx \frac{-2.3026}{-0.6931} \approx 3.32$$

So they need at least **4 bits**.

Answer: $\boxed{4 \text{ bits}}$

Final Answers

(a) 0%
(b) 50%
(c) 4 bits